

Introduction to Arduino (Part 2)

The goal of this TP is to add a few functionalities to the night detector device. We will add a push button to change the state of the light to either AUTO, ON, or OFF, and a potentiometer to control the brightness of the LED.

Important :

This lab involves using electronic equipment and wiring them together. Incorrect wiring may cause short circuits that may damage the boards. Make sure:

1. Always disconnect the USB port of the Arduino board when wiring.
2. In the first labs, always call the professor to verify the wiring before putting the power on by connecting the USB cable.

Evaluation :

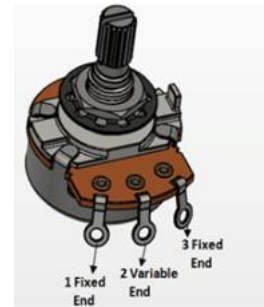
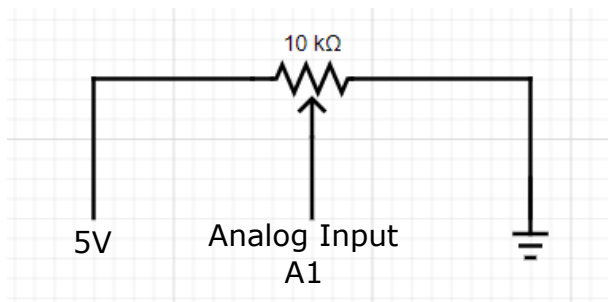
These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Report: at the end of the class, follow the instructions of the assignment section of the TP, and send your report to your TP teacher **before midnight the day of the TP**:

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The TP must be done in groups of students. The groups must stay the same for all the next TPs. When you send your code, put the following subject on the email « [IoT] TP1-P2-Group <put your names here> ».

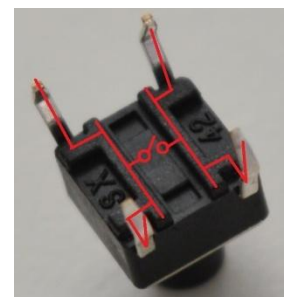
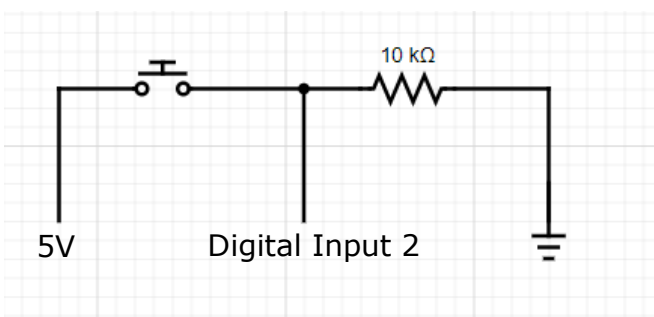
- TP 1 : Introduction to Arduino
- Professor: Ouassim Karrakchou



2. Call the Professor.
3. Add the code to measure the value of your potentiometer and display it on the Serial monitor to verify that it changes when you turn the potentiometer.
4. You will notice that the measured value ranges from **0 to 1023**. We want to use **PWM** to control the **brightness** of the LED. From the course, you know that for PWM, we need a value between 0 and 255. We thus need to use a "règle de trois" to convert the measured potentiometer value to a value between **0 and 255**.
For this, you need to use the **map** function from the Arduino framework (<https://www.arduino.cc/reference/en/language/functions/math/map/>). Once you convert the value to the range 0-255, display the new value on the Serial monitor.
5. Call the professor.
6. Use **analogWrite** instead of **digitalWrite** to control the brightness of the LED **when the LED is supposed to be ON**.
7. Call the Professor.

III- Add the push button to change the state of the LED:

1. Disconnect your board and connect the push button based on the following circuit:



2. Call the Professor.

- #### IV- Use an interrupt for controlling the state of the LED:

- ```
enum LED_STATE {
 AUTO,
 ON,
 OFF
};

LED_STATE state = AUTO;

void changeStatus() {
 switch (state) {
 case AUTO:
 state = ON;
 Serial.println("We switch to the Always ON state");
 break;
 case ON:
 state = OFF;
 Serial.println("We switch to the Always OFF state");
 break;
 case OFF:
 state = AUTO;
 Serial.println("We switch to the Auto state");
 break;
 }
}
```

2. Add to your **setup()** function the code needed to call **changeState** whenever the button is pressed. You need to use **attachInterrupt** from the Arduino framework (make sure you use the **digitalPinToInterrupt(pin)** syntax):  
<https://www.arduino.cc/reference/en/language/functions/external-interrupts/attachinterrupt/>. The mode needs to be **RISING** so that the **changeState** function is called when the button is pressed (i.e., its pin goes from LOW to HIGH). **You'll also need to remove the changes done in the previous part (digitalRead...).**
3. Call the Professor.
4. Modify your **loop()** function accordingly to check the state variable, and based on its value, either keep the LED ON, OFF, or do the automatic change based on the luminosity
5. Call the Professor.

Write a report showing the different steps you did in the TP to create your updated night detector IoT device. Use screenshots and attach the code you used.